

Vaishnavi Tompe

Full Stack AI Engineer

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SUMMARY

Full Stack AI Engineer with 2.5+ years of experience building production-grade AI systems using LLMs, RAG pipelines, and multi-agent architectures. Skilled in FastAPI, React, Azure OpenAI, LangChain, and LangGraph. Experienced in designing end-to-end Generative AI systems, including retrieval pipelines, prompt optimization, token cost reduction, and LLM evaluation workflows. Strong focus on scalable AI architecture and real-world deployment.

SKILLS

- Programming Languages: Python, JavaScript, TypeScript
- Frontend: React.js, Next.js, Redux, HTML5, CSS3, Bootstrap
- Backend: Node.js, Express.js, FastAPI
- GenAI / AI Technologies: Azure OpenAI, LangChain, LangGraph, Vector Databases (FAISS, Pinecone), Embeddings (OpenAI), LLM Orchestration, Multi-Agent Systems, Retrieval Pipelines, Token Optimization, Prompt Optimization
- Databases: MongoDB, SQL
- Cloud & DevOps: Azure, Docker, CI/CD, Vercel, Render
- Tools: Git, GitHub, Postman

EXPERIENCE

Full Stack AI Engineer – GenAIKit Pvt Ltd

(Jan 2024 – June 2026)

- Built and deployed BharatShodh (bharatshodh.com), a multilingual Indian-language chatbot on Azure VM.
- Designed hybrid RAG pipelines (FAISS + OpenAI embeddings) for large-scale retrieval, improving answer accuracy by 35% on production queries.
- Designed and deployed scalable RESTful APIs using FastAPI and Node.js, improving system performance by 30%.
- Built and orchestrated **LLM-based** workflows and multi-agent systems using LangGraph, enabling automated task execution and reducing manual effort by 40%.
- Optimized **prompt engineering** and token usage strategies, reducing API costs and improving response accuracy and latency.
- Implemented secure **authentication** and **authorization** mechanisms, ensuring data protection and reliable user access control.
- Implemented LLM evaluation and monitoring workflows using tracing and performance metrics to improve response quality.
- Worked with structured data and LLM-based query generation pipelines

Intern – GenAIKit Pvt Ltd

(Oct 2023 – Dec 2023)

- Built LLM-powered applications using RAG pipelines
- Developed data ingestion to response generation workflows
- Worked on prompt optimization and agent workflows

PROJECTS

AI Interview Assistant

Technology: FastAPI backend + LangChain orchestration + vector database + Azure OpenAI

- Built real-time voice-based AI assistant
- Integrated speech-to-text and text-to-speech
- Implemented RAG pipeline for contextual answers

RAG Q&A Platform

Technology: FastAPI backend + LangChain orchestration + vector database + Azure OpenAI

- Developed retrieval-based system for financial datasets
- Implemented vector search using FAISS
- Improved query relevance and reduced hallucination

Client Web Application

Independent Freelance Project

- Integrated Azure OpenAI for intelligent response generation in healthcare workflow
- Collaborated directly with client to deliver production-ready features, enhancing checkout reliability and customer experience.

ROLES AND RESPONSIBILITY

- Developed and maintained end-to-end Full Stack AI applications using React.js, FastAPI, Node.js, MongoDB, and Azure OpenAI.
- Designed and implemented RAG architectures leveraging **LangChain**, **LangGraph**, vector databases, embeddings, and semantic search techniques.
- Built scalable backend APIs, microservices, and AI orchestration workflows to support **production-grade Generative AI** solutions.
- Integrated **LLMs** into business applications, optimizing prompts, token usage, response quality, and system performance.
- Developed interactive and responsive frontend applications with React.js, ensuring seamless integration with **AI-powered** backend services.
- Implemented authentication, authorization, monitoring, logging, and security best practices for enterprise-grade deployments.
- Deployed and managed AI solutions on Azure using **Docker**, CI/CD pipelines, and cloud infrastructure while ensuring scalability and reliability.

EDUCATION

Bachelors of Technology - Computer Science and Engineering

MGM's College of Engineering

CERTIFICATION

Full Stack Developer